Looking at the quantum finance simulation code, I'll provide a rigorous mathematical definition of the stochastic process underlying David Orrell's quantum oscillator model.## Mathematical Definition of the Quantum Oscillator Stochastic Process

1. State Space and Basic Setup

Let S_t denote the asset price at time t, and define the log-price process:

$$X_t = \log S_t$$

The process evolves through discrete quantum jumps at times $\{t_k\}_{k=0}^{\infty}$ where $t_{k+1} - t_k = \Delta t$.

2. Quantum State Representation

The system is described by a quantum harmonic oscillator with:

- **Energy levels**: $n \in \{0,1,2,\dots,n_{\max}\}$
- **Wave functions**:

$$\psi_n(x) = N_n H_n\left(\frac{x}{\alpha}\right) \exp\left(-\frac{x^2}{2\alpha^2}\right)$$

where:

- H_n is the n-th Hermite polynomial
- $\alpha = \sqrt{\sigma\sqrt{\Delta t}}$ is the scaling parameter
- $N_n = \frac{1}{\sqrt{2^n n! \sqrt{\pi}\alpha}}$ is the normalization constant

3. Energy Level Distribution

The probability of the system being in energy level n follows a thermal distribution:

$$P(n) = \frac{e^{-n/T}}{\sum_{k=0}^{n_{\max}} e^{-k/T}}$$

where T is the market "temperature" parameter.

4. The Q-Distribution

The probability density for a price change Δx is given by the **q-distribution**:

$$f_Q(\Delta x) = \sum_{n=0}^{n_{\max}} P(n) \cdot |\psi_n(\Delta x)|^2$$

This is a weighted superposition of squared quantum wavefunctions, which can be written explicitly as:

$$f_Q(\Delta x) = \sum_{n=0}^{n_{\max}} P(n) \cdot N_n^2 \cdot H_n^2\left(\frac{\Delta x}{\alpha}\right) \cdot \exp\left(-\frac{\Delta x^2}{\alpha^2}\right)$$

5. The Jump Process

The log-price process evolves as a pure jump process:

$$X_{t+\Delta t} = X_t + J_t$$

where J_t is a quantum jump with distribution:

$$J_t \sim f_Q(\cdot)$$

6. Full Stochastic Process Definition

The complete stochastic process can be defined as:

$$dX_t = dN_t \cdot J_t$$

where:

- N_t is a Poisson process with intensity $\lambda = 1/\Delta t$
- J_t are i.i.d. random variables distributed according to f_Q
- The jumps J_t are independent of the Poisson process N_t

7. Conditional Variance Structure (Q-Variance)

The key property of this process is that the conditional variance over period T given a price change x follows:

$$\mathrm{Var}[X_{t+T} - X_t \mid X_{t+T} - X_t = x] = \sigma^2 + \frac{z^2}{2}$$

where $z = \frac{x}{\sqrt{T}}$.

8. Moments and Characteristic Function

The characteristic function of the increment $\Delta X_t = X_{t+\Delta t} - X_t$ is:

$$\phi_{\Delta X}(u) = \mathbb{E}[e^{iu\Delta X}] = \sum_{n=0}^{n_{\max}} P(n) \cdot \phi_n(u)$$

where $\phi_n(u)$ is the Fourier transform of $|\psi_n(x)|^2$:

$$\phi_n(u) = \int_{-\infty}^{\infty} e^{iux} |\psi_n(x)|^2 dx$$

9. Key Properties

The process exhibits:

1. **Heavy tails**: The kurtosis exceeds 3 due to the Hermite polynomial structure
$$\mathrm{Kurt}[\Delta X_t] > 3$$
2. **Scale invariance**: The q-variance relationship holds for all time scales
$$\mathrm{Var}[\Delta X_T | \Delta X_T = x] = \sigma^2 + \frac{x^2}{2T} \ \forall T > 0$$
3. **Minimum volatility**: Even at $n = 0$, there is non-zero variance
$$\mathrm{Var}[\Delta X_t] \geq \sigma^2 \Delta t > 0$$
4. **Discrete energy quantization**: Price changes occur through transitions between discrete energy levels

10. Comparison with Classical Processes

Unlike classical processes:

- **Brownian motion**: $dX_t = \mu dt + \sigma dW_t$ (continuous, Gaussian)
- **Jump-diffusion**: $dX_t = \mu dt + \sigma dW_t + J dN_t$ (mixed continuous-jump)

The quantum process is:

- **Pure jump**: No continuous component
- **Non-Lévy**: Does not satisfy infinite divisibility
- **Quantum superposition**: Jumps are drawn from a superposition of quantum states
- **Energy-dependent**: Jump size distribution depends on quantum energy levels

11. Entropic Force Interpretation

The process can be viewed as driven by entropic forces:

$$F(x) = -\frac{\partial V(x)}{\partial x}$$

where $V(x) = \frac{1}{2} m\omega^2 x^2$ is the harmonic potential, with:

- m representing market "inertia"
- $\omega = \sigma/\sqrt{\Delta t}$ as the oscillator frequency

This provides a physical interpretation where price movements are resisted by entropic forces that increase with displacement from equilibrium.